Xiaohan Zou

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Education

Pennsylvania State University — State College, PA
Ph.D. in Computer Science and Engineering — 08/2023 - 05/2027 (Expected)
Advisors: Prof. George Kesidis & Prof. Lu Lin

Boston University — Boston, MA
M.S. in Computer Science — 09/2021 - 01/2023
Advisor: Prof. Bryan Plummer

Tongji University — Shanghai, China
B.Eng. in Software Engineering — 09/2016 - 07/2020

Selected Papers (also see Google Scholar)

Publications

P1 **Understanding and Rectifying Safety Perception Distortion in VLMs**. Xiaohan Zou, Jian Kang, George Kesidis, Lu Lin. *NeurIPS 2025*. [paper]

P2 **Learning Conditional Space-Time Prompt Distributions for Video Class-Incremental Learning**. Xiaohan Zou, Wenchao Ma, Shu Zhao. *CVPR 2025, **Highlight***. [paper]

P3 **Leveraging an LLM-Enhanced Bilingual Conversational Agent for EFL Children's Dialogic Reading: Insights from Children, Parents, and Educators**. Feiwen Xiao, Zhaohui Li, Jiaju Lin, Xiaohan Zou, Dandan Yang, Ellen Wenting Zou, Jinjun Xiong. *Computers and Education: Artificial Intelligence, 2025*.

P3 **Efficient Meta-Learning for Continual Learning with Taylor Expansion Approximation**. Xiaohan Zou, Tong Lin. *IJCNN 2022, **Oral***. [paper]

P4 **To be an Artist: Automatic Generation on Food Image Aesthetic Captioning**. Xiaohan Zou, Cheng Lin, Yinjia Zhang, Qinpei Zhao. *ICTAI 2020, **Oral***. [paper]

P5 **A Survey on Application of Knowledge Graph**. Xiaohan Zou. *CCEAI 2020*. [paper]

Preprints

R1 **When Vision-Language Models Judge Without Seeing: Exposing Informativeness Bias**. Xiaohan Zou, Roshan Sridhar, Mohammadtaher Safarzadeh, Dan Roth. *2025*.

R2 **Reconstruct before Query: Continual Missing Modality Learning with Decomposed Prompt Collaboration**. Shu Zhao, Xiaohan Zou, Tan Yu, Huijuan Xu. *2024*. [paper]

R3 **TokenFlow: Rethinking Fine-grained Cross-modal Alignment in Vision-Language Retrieval**. Xiaohan Zou, Changqiao Wu, Lele Cheng, Zhongyuan Wang. *2022*. [paper]

Industry Experience

Oracle AI — Redwood, CA
GenAI Research Intern — 05/2025 - 08/2025
Topic: Toward More Reliable VLM-as-a-Judge — *Submitted to ACL Rolling Review (R1)*

- Revealed that Vision-Language Models (VLMs), when used as judges, often simply favor more informative answers while overlooking inconsistencies with the image, even when they can recognize them.
- Developed a simple yet effective judging paradigm that mitigates over-prioritization of informativeness and refocuses attention on the image.

Kuaishou Technology Beijing, China
Machine Learning Engineer Intern 07/2021 - 12/2021
Topic: Video-Text Retrieval *Technical Report (R3)*

- Proposed a unified framework for image- and video-text retrieval and enabled finer-grained cross-modal semantic alignment through optimal transport within this framework.
- Created a PyTorch codebase to streamline video-text retrieval for both research and production use.

Research Projects

Safety Perception Distortion in VLMs *Accepted at NeurIPS 2025 (P1)*
Pennsylvania State University State College, PA

- Demonstrated how visual modality shifts activations of unsafe vision-language inputs to a seemingly safer space, ultimately distorting the safety perception of Vision-Language Models (VLMs).
- Proposed disentangling modality shifts and removing safety-relevant components to reactivate VLMs' defense mechanisms while maintaining helpfulness and utility.

Video Class-Incremental Learning *Published at CVPR 2025 (P2)*
Pennsylvania State University State College, PA

- Modeled the prompt space of CLIP as a conditional distribution with a diffusion model for video class-incremental learning, improving generalization and achieving SoTA results.
- Developed a prompting strategy that enables CLIP to capture spatio-temporal relationships in videos.

Continual Missing Modality Learning *Technical Report (R2)*
Pennsylvania State University State College, PA

- Introduced a new task for handling non-static data with missing modalities, along with a new benchmark.
- Decomposed prompts into modality-specific components and utilized knowledge from pre-trained multimodal models to reconstruct inputs for absent modalities.

Awards & Honors

Bronze, China Collegiate Programming Contest (CCPC) 2018
Second Prize, China Mathematical Contest in Modeling (CUMCM) 2017, 2018
Second Prize, Tongji University Programming Contest 2017, 2018
Second Prize, East China Normal University Programming Contest 2017

Professional Activities

Conference Reviewer for AAAI, NeurIPS, CVPR, ICLR, AIED, BigData
Vice Chief Technology Officer & Chief Experience Officer at Tongji Microsoft Student Club 2018 - 2019

Teaching Assistant

PSU CMPSC 360 Discrete Mathematics Fall 2025, Spring 2025
PSU CMPSC 465 Data Structures and Algorithms Fall 2024

Skills

Programming Languages: Python, JavaScript / TypeScript, HTML / CSS, Java, C++, MATLAB
Tools and Frameworks: PyTorch, , Transformers, Vue, React, Flask, Linux, Git, LaTeX
Languages: Chinese (native), English (proficient)